

PET-to-Filament Recycling

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Abstract— The increasing consumption of polyethylene terephthalate (PET) bottles has led to severe environmental concerns due to plastic waste accumulation. Recycling PET into 3D printing filament offers a sustainable approach for waste reduction and distributed manufacturing. This paper presents the design and development of a compact PET-to-filament recycling system aimed at small laboratories, educational institutions, and makerspaces. The proposed system converts post-consumer PET bottles into usable 1.75 mm filament through bottle stripping, controlled extrusion, cooling, and spooling. An ESP32-based control system with a web-based interface enables precise temperature and pull-speed regulation. Initial extrusion trials demonstrate the feasibility of producing consistent filament suitable for mechanical testing and functional 3D printing applications. Challenges related to moisture sensitivity, diameter consistency, and material uniformity are discussed, along with ongoing efforts toward mechanical characterization using tensile, compressive, and bending tests. **Keywords**— PET recycling, 3D printing filament, sustainable manufacturing, plastic waste, extrusion, ESP32 control

1. Introduction

Polyethylene terephthalate (PET) is one of the most widely used plastics in consumer packaging, particularly for beverage bottles. Despite being highly recyclable, a significant portion of PET waste ends up in landfills due to inefficiencies in centralized recycling systems. Recently, decentralized recycling methods have gained attention, especially the conversion of waste PET into 3D printing filament for additive manufacturing applications.

PET-to-filament recycling involves washing used bottles, removing labels, cutting the bottles into strips, and melting the material at elevated temperatures (typically 250–260°C) in an extrusion system. The extruded material is cooled and pulled into filament with a controlled diameter. Recycled PET filament is known to produce strong and stiff printed parts suitable for functional and mechanical applications, although challenges such as moisture absorption, color inconsistency, and viscosity variation remain.

This work focuses on developing a compact, low-cost PET recycling system that enables closed-loop fabrication and improves accessibility for small-scale users.

A. Abbreviations and Acronyms

Abbreviation	Full Form	First-use Context
PET	Polyethylene Terephthalate	Material used for recycling
PLA	Polylactic Acid	Reference filament material
ABS	Acrylonitrile Butadiene Styrene	Reference filament material
ESP32	Espressif Systems 32-bit Microcontroller	Control system
UTM	Universal Testing Machine	Mechanical testing
UI	User Interface	Web-based control
NEMA	National Electrical Manufacturers Association	Stepper motor standard
CAD	Computer-Aided Design	Mechanical design
RPM	Revolutions Per Minute	Spooler speed
PID	Proportional–Integral–Derivative	Temperature control (if implied)
FDM	Fused Deposition Modeling	3D printing process
AM	Additive Manufacturing	Manufacturing context

A. Units

Symbol	Meaning	Unit
T	Extrusion temperature	°C
d	Filament diameter	mm
v	Pull speed	mm/s
σ	Stress	MPa
ϵ	Strain	–
F	Applied force	N
A	Cross-sectional area	mm ²
L	Gauge length	mm
ω	Spooler angular speed	rad/s

B. Equations

The mechanical and processing behaviour of the recycled PET filament was quantified using standard analytical relations. The tensile stress developed during testing was calculated from the applied load and original cross-sectional area using

$$\sigma = \frac{F}{A} \quad (1)$$

where σ is the tensile stress (MPa), F is the applied force (N), and A is the original cross-sectional area (mm²). The corresponding tensile strain was obtained from the change in gauge length according to

$$\varepsilon = \frac{\Delta L}{L_0} \quad (2)$$

where ε is the strain, ΔL is the elongation (mm), and L_0 is the initial gauge length (mm). Young's modulus of the recycled PET filament was determined from the slope of the linear elastic region of the stress-strain curve using

$$E = \frac{\Delta \sigma}{\Delta \varepsilon} \quad (3)$$

where E is Young's modulus (GPa). Filament diameter stability was evaluated by calculating the deviation from the nominal filament diameter of 1.75 mm, expressed as

$$\Delta d = d_{\text{measured}} - d_{\text{nominal}} \quad (4)$$

where d_{measured} is the measured filament diameter and d_{nominal} is the nominal diameter. The extrusion process was further analyzed using the mass continuity relation to correlate filament pull speed with material flow rate, given by

$$\dot{m} = \rho A v \quad (5)$$

where $\dot{m}$ is the mass flow rate (kg/s), ρ is the density of PET (kg/m³), A is the filament cross-sectional area (m²), and v is the filament pull speed (m/s).

2. Problem Definition and Challenges

2.1 Expensive and Bulky Systems

Existing PET recycling setups typically consist of multiple machines such as shredders, washers, dryers, extruders, and spoolers. These systems are bulky, expensive, and unsuitable for small laboratories or individual users.

2.2 Quality Control Challenges

Maintaining consistent filament quality is difficult due to variations in flake or strip size, moisture content, and inadequate melt filtration. These issues often lead to brittle filament, diameter fluctuations, and poor printability.

2.3 Limited Accessibility

Although professional PET recycling systems exist, their high cost restricts widespread adoption, particularly in educational and low-resource settings.

3. Objectives

3.1 Main Objective

To design and develop a compact system capable of converting waste PET bottles into usable 3D printing filament.

3.2 Sub-Objectives

- To test the mechanical strength of recycled PET filament in comparison with PLA and ABS through tensile, compressive, and bending tests.
- To develop an ESP32-based controller with a fully functional web-based user interface.
- To implement a high-torque NEMA17-driven spooler system.
- To achieve stable filament diameter and continuous filament pull during extrusion.

4. Methodology

4.1 Experimental Plan

The PET recycling process follows these steps:

1. Stripping PET bottles into continuous plastic strips using a bottle cutter.
2. Feeding the strips into a heated extrusion system.
3. Calibrating extrusion temperature and filament pull speed to ensure stable flow.
4. Cooling the extruded filament and spooling it at a controlled rate.

4.2 Instruments and Software

- Universal Testing Machine (UTM) for mechanical characterization
- ESP32 microcontroller for system control
- Locally hosted web interface for real-time parameter adjustment
- 3D-printed PLA components
- Temperature sensors and stepper motor drivers

5. Experimental Setup

The experimental setup consists of a PET bottle strip cutter, a heated extruder, a cooling section, and a motor-driven spooler. The extruder melts PET at temperatures between 250–260°C, after which the molten polymer is cooled and pulled into filament. The spooler speed is controlled using the ESP32-based system to maintain a consistent filament diameter. The produced filament is

intended for use in standard 3D printers, followed by mechanical testing of printed specimens using a UTM.

6. Materials and Equipment

Materials:

- Post-consumer PET bottles
- PLA printed structural components
- Mechanical hardware

Equipment:

- Extruder assembly
- Spooler mechanism
- ESP32 control module
- PET bottle strip cutting device

7. Results and Discussion

Initial trials focused on evaluating filament diameter consistency, extrusion stability, and visual characteristics of the produced filament. Preliminary results indicate that the system is capable of producing hollow PET filament with acceptable dimensional accuracy. Temperature control and extrusion stability were successfully validated, while spooler speed was fine-tuned using the web-based ESP32 interface. Variations in filament clarity and diameter were observed, highlighting the importance of precise moisture control and pull-speed regulation.

8. Project Status

8.1 Work Completed

- Mechanical design and fabrication of extruder mounts, spooler, and PLA components
- Fabrication of PET bottle strip cutting tool
- Initial PET filament extrusion trials
- Partial integration of mechanical and electronic components on a base structure

8.2 Work Remaining

- Detailed filament diameter consistency testing
- Printing standardized test specimens using recycled PET filament
- Mechanical testing (tensile, compressive, and bending tests) using UTM
- Stress-strain analysis and comparative study with PLA and ABS

9. Conclusion

This work demonstrates the feasibility of a compact PET-to-filament recycling system for decentralized and sustainable manufacturing. Initial results show promising extrusion stability and controllability, with ongoing efforts directed toward improving filament consistency and mechanical performance. The proposed system has the potential to reduce plastic waste while enabling low-cost production of functional 3D printing filament.

C. Figures and Tables



Figure 2 PET bottle strip cutting setup

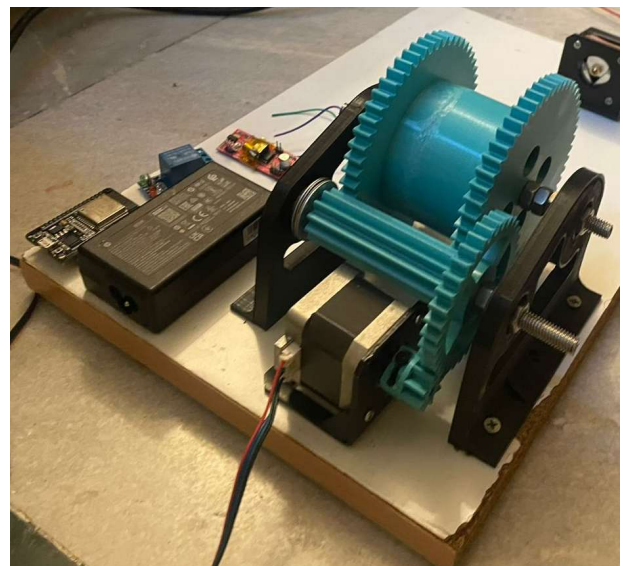


Figure 1 Machine parts and components used

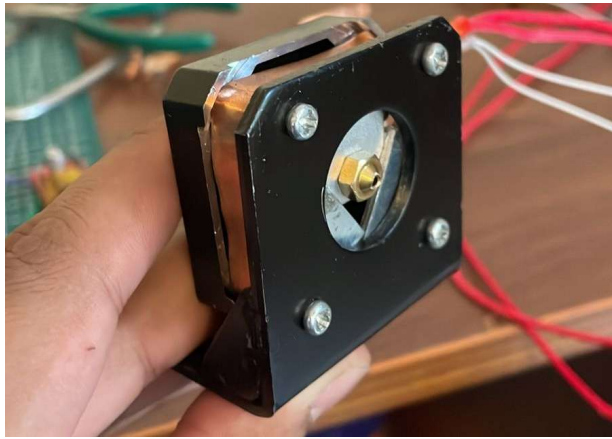


Figure 3 Extruder nozzle modified to 1.75mm

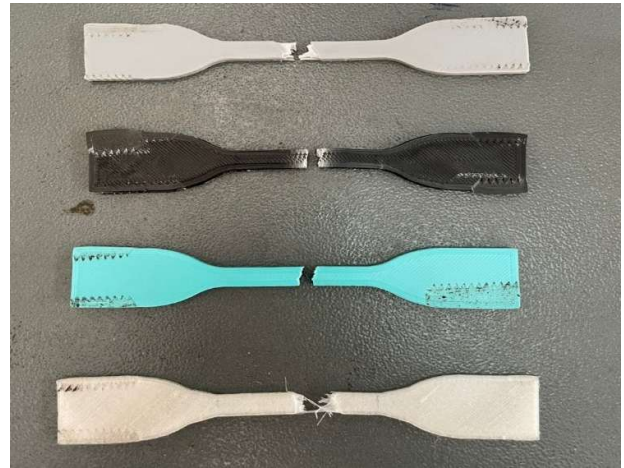


Figure 6 PETG, ABS, PLA and PET specimen

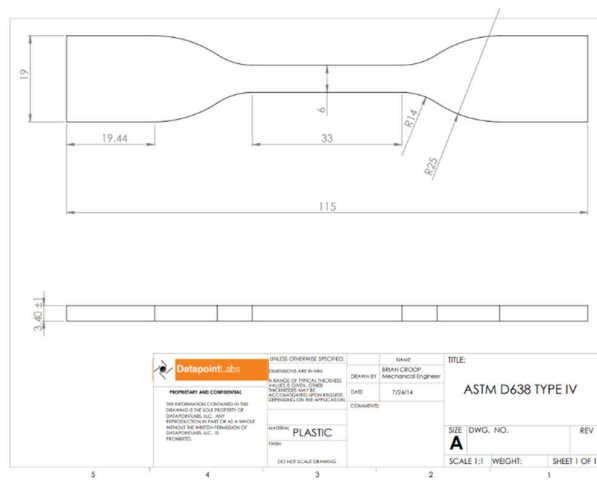


Figure 4 ASTM Specimen drawing

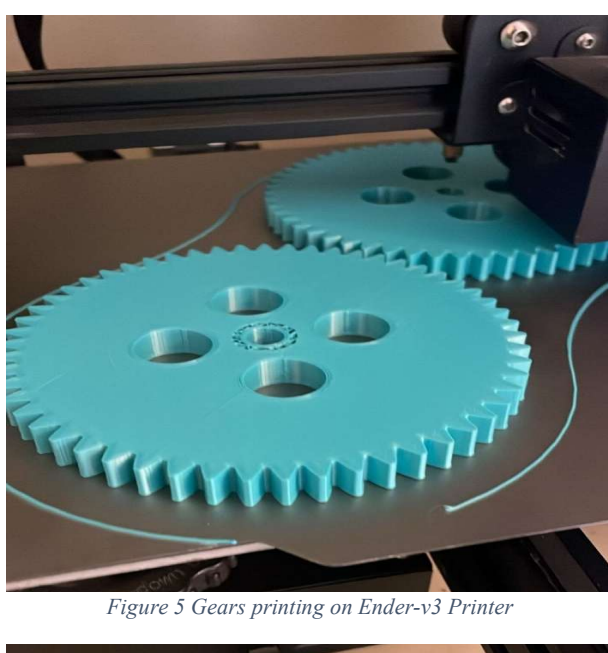


Figure 5 Gears printing on Ender-v3 Printer

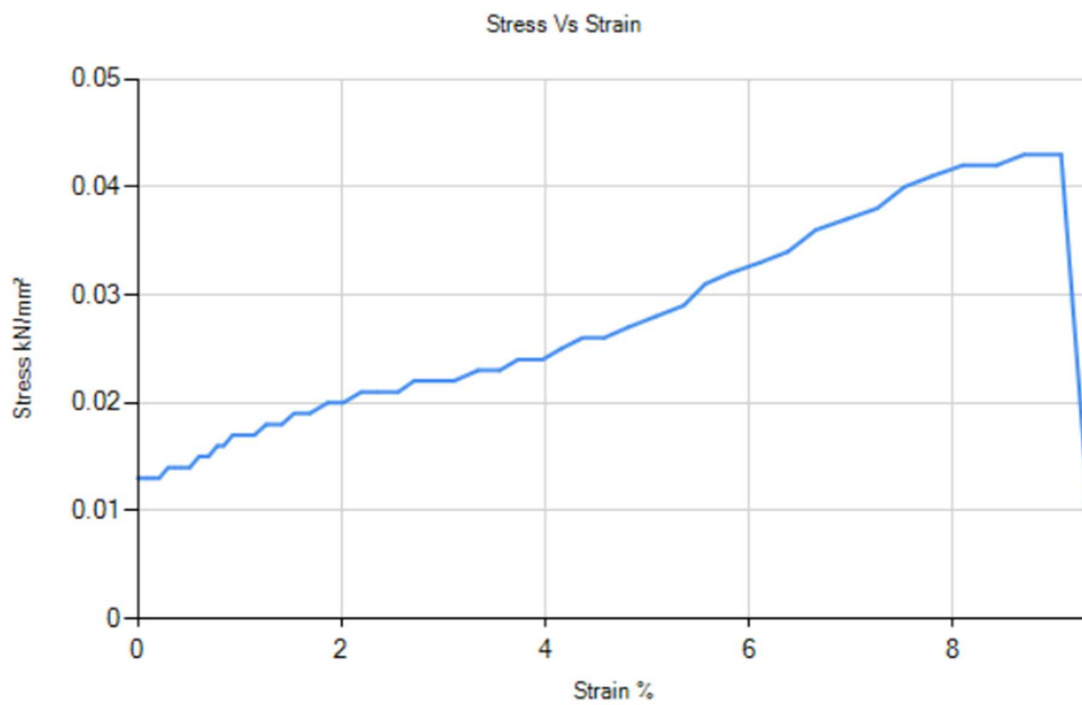


Figure 7 PLA stress-strain curve

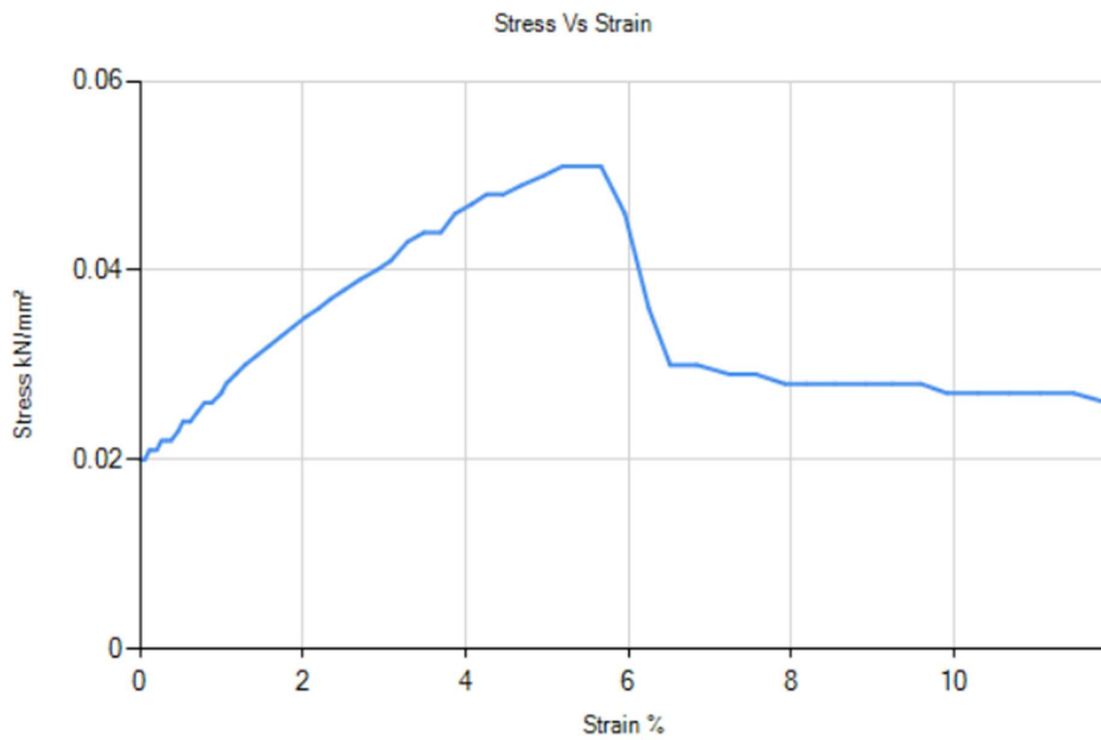


Figure 8 PETG stress-strain curve

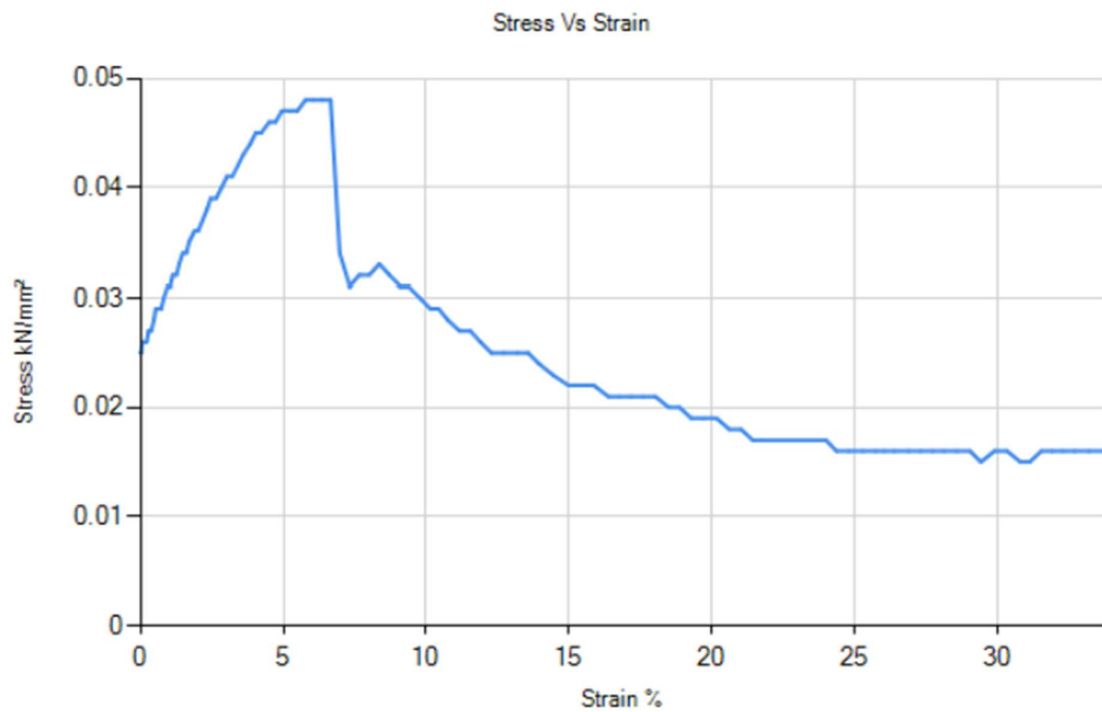


Figure 10 PET stress-strain curve

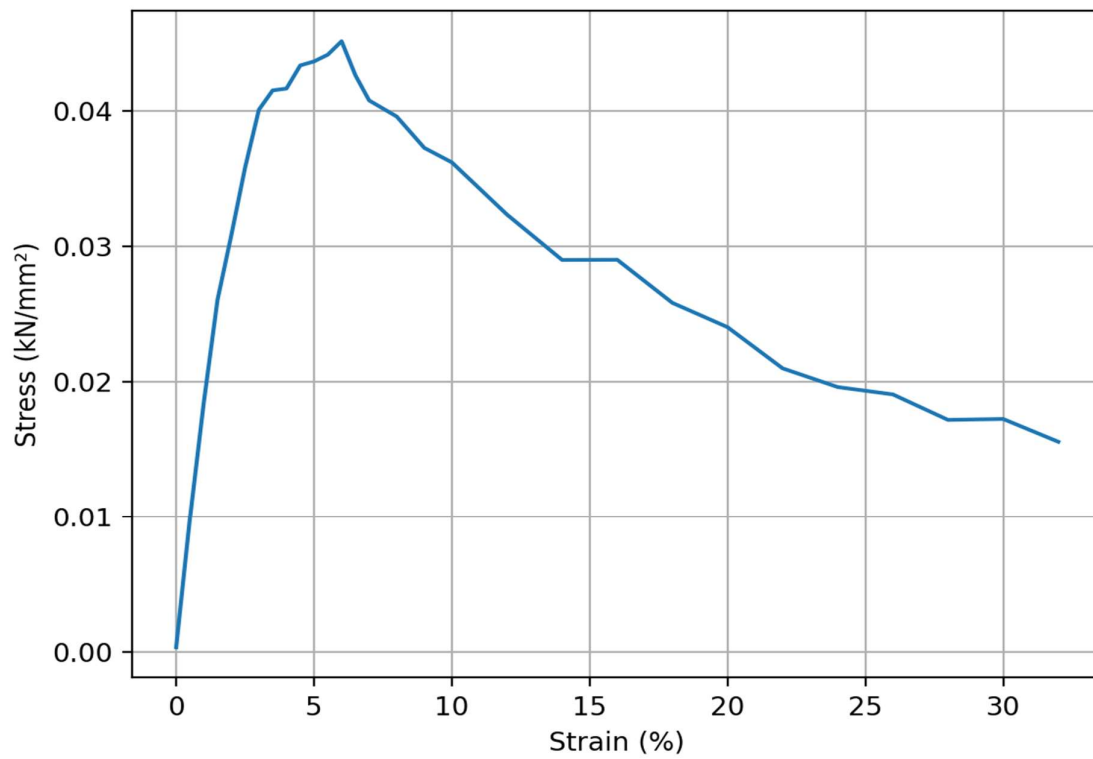


Figure 9 ABS stress-strain curve

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